Application Data Sheet 37 CFR 1.76

The Application Data Sheet is part of the provisional or non-provisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76.

Inventor Information

of inventors: 1

1. **Eric Haywood**

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Application Information

Customer number	**221639 -**
Correspondence address	---
Title of invention	**AI SESSION REPLAY ENGINE WITH DUAL-PANE TEMPORAL SCRUBBING FOR AGENTIC EXECUTION FORENSICS**
Attorney docket number	**---**
Entity status	---

Application type	Provisional Application under 35 USC 111(b)
Subject matter	Utility
Total number of drawing sheets	---
Filing by reference	No

Representative Information

of representatives: 0

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32).

Data was not provided for this section.

Domestic Benefit/National Stage Information

of benefit claims: 0

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c), 386(c), or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.

Data was not provided for this section.

Foreign Priority Information

of foreign priority claims: 0

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the Application Data Sheet constitutes the claim for priority as required by 35 U.S.C. 119 (b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX) the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a

copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Data was not provided for this section.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

Checking this box will cause the application to be examined under the first inventor to file provisions of the AIA.

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 2016, 2013, will be examined under the first inventor to file provisions of the AIA.

Authorization or Opt-Out of Authorization to Permit Access

When this Application Data Sheet is properly signed and filled with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE:
This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. **Priority Document Exchange (PDX)**
Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the China National Intellectual Property Administration (CNIPA), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its

related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h) (1).

B. **Search Results from U.S. Application to EPO**

Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with the search results from the instant application.

NOTE:

Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Applicant Information

of applicants: 0

The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46.

Data was not provided for this section.

Assignee Information including Non-Applicant Assignee Information

of assignees: 1

An assignee-applicant identified in the "Applicant" section will appear on the patent application as an applicant.

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

HeadySystems Inc.
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Organization

Signature

NOTE:

This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). **However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c)**

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See CFR 1.4(d) for the manner of making signatures and certifications.

Signature	First name	Last name	Registration #	Date
/Eric Haywood/	ERIC	HAYWOOD	---	03/03/2026

UNITED STATES
PATENT AND TRADEMARK OFFICE

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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
63/995,318	**03/03/2026 02:10:10 PM Z ET**	**-**

Title of Invention

AI SESSION REPLAY ENGINE WITH DUAL-PANE TEMPORAL SCRUBBING FOR AGENTIC EXECUTION FORENSICS

Application Information

APPLICATION TYPE	Utility - Provisional Application under 35 USC 111(b)	PATENT #	-
CONFIRMATION #	7203	FILED BY	ERIC HAYWOOD
PATENT CENTER #	74697235	FILING DATE	-
CUSTOMER #	221639	FIRST NAMED INVENTOR	Eric Haywood
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	-

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
generatedADS74697235.pdf	5	Application Data Sheet	109 KB
HS-2026-006-Final-Assigned-Updated-APP.TEXT.docx	4	Application body structured text document	39 KB

Warning: One or more pages are missing page numbering. Page numbering will be automatically applied after submission.

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
generatedADS74697235.pdf	49D8D0F6B91BE7D47DC676ED2651646D53796EF27E3474663

3DC5A7225D788F078E1AE93A2662F58819736BE130A66BFFE
ACF5D377C8BBCB5C7C51F5524E4E10

HS-2026-006-Final-Assigned-Updated-APP.TEXT.docx

3B7AD9756C6109BF9CDA5660CD7F96C0510CE84B19B0E1FA
6979FE9188FB5D5C7E014546E579D131BB22310059B27C4C9
15B8EEDB37F3C50B4E134AFC9A65C53

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PROVISIONAL PATENT APPLICATION

Under 35 U.S.C. § 111(b)

AI SESSION REPLAY ENGINE WITH DUAL-PANE TEMPORAL SCRUBBING FOR AGENTIC EXECUTION FORENSICS

Title of the Invention	AI Session Replay Engine with Dual-Pane Temporal Scrubbing for Agentic Execution Forensics
Docket Number	HS-2026-006
Inventor(s)	Eric Haywood
Residence(s)	149 Remington Street Unit 425, Fort Collins, CO 80524
Correspondence Address	HeadySystems Inc., 149 Remington Street Unit 425, Fort Collins, CO 80524
Attorney/Agent (Name & Reg. No.)	None
U.S. Government Interest (Yes/No)	No
Gov. Agency & Contract No.	N/A

Title of the Invention

[0001] AI Session Replay Engine with Dual-Pane Temporal Scrubbing for Agentic Execution Forensics

Cross-Reference to Related Applications

[0002] This application claims the benefit of priority and is related to co-pending provisional applications filed concurrently herewith by the same inventor and assignee.

Field of the Invention

[0003] This invention relates to systems and methods for recording, indexing, and replaying AI agent execution sessions with frame-level granularity across multiple concurrent interaction channels.

Background of the Invention

[0004] Current AI agent systems lack the ability to perform forensic analysis of past execution sessions. When an AI agent makes decisions, there is no standardized mechanism to rewind and observe the exact sequence of events—both the user-facing outputs and the internal agent decision waterfall—that led to a specific outcome. Traditional logging provides text-based audit trails but lacks temporal precision, dual-pane visualization, and interactive scrubbing capabilities analogous to video DVR controls.

Summary of the Invention

[0005] The invention provides a system and method as described herein.

[0006] Frame Recording: Each discrete event (user input, agent decision, API call, state mutation, UI render) is captured as an indexed frame with sub-millisecond timestamps and stored sequentially.

[0007] Dual-Pane Replay: Sessions are replayed in a synchronized dual-pane view showing (a) the user-facing UI state and (b) the internal agent execution waterfall, enabling precise temporal scrubbing.

[0008] Columnar Storage: Frames are stored in a columnar database to enable efficient time-range queries and aggregation. Columnar storage compresses data and processes columns together for high-performance analytics.

[0009] Session Search: Sessions are searchable by user ID, metadata, temporal range and decision outcome patterns, supporting full forensic analysis.

[0010] Frame-Level Indexing: Each frame maintains references to the agent decision tree node, enabling root-cause analysis of any output.

Detailed Description of the Preferred Embodiment

A. SessionRecorder

[0011] The SessionRecorder component maintains an in-memory ring buffer of a configurable number of sessions. For each session it records frames containing a frame index, sub-millisecond timestamp, event type (user input, agent decision, API call, state mutation, UI render, error or system event), structured payload data, the agent decision tree path from root to current node, a serialized UI state snapshot, and a delta encoding of the state changes for efficient storage.

[0012] The recorder writes these frames to columnar storage (for example, ClickHouse) in near real time. Columnar storage organizes data by column rather than by row, allowing compression and efficient range scans and thus enabling high-performance time-range queries.

B. Replay Engine

[0013] The Replay Engine reconstructs any recorded session on demand. Given a session identifier and an optional frame range, it rebuilds cumulative UI state from the initial state plus sequential deltas, aligns agent waterfall events to corresponding UI state changes, and produces a synchronized dual-pane output that supports forward and backward temporal navigation.

C. Search and Analytics

[0014] The system provides full-text and structured search across all recorded sessions. Users may filter by user identifier, device type, browser, geographic region, temporal range, and specific decision-path patterns. It also generates per-session analytics such as frame count, duration, number of agent decisions, error rate and dominant decision paths.

Claims

1. A computer-implemented method for recording an AI agent execution session, comprising: (a) capturing each discrete event during the session as an indexed frame containing a timestamp, event type, payload, agent decision tree path and UI state; (b) storing the indexed frames in a columnar database with sub-millisecond temporal precision; and (c) providing a replay interface enabling temporal scrubbing of the recorded session in a dual-pane view simultaneously showing user-facing UI state reconstruction and internal agent decision waterfall.

2. The method of claim 1, wherein the UI state in each frame is stored as a delta from the previous frame, reducing storage requirements while maintaining the ability to reconstruct full state at any temporal position.

3. The method of claim 1, further comprising generating a session summary including frame count, session duration, agent decision count, error rate, and dominant agent decision path.

4. A system for AI agent forensic analysis comprising: a session recorder that captures indexed frames during agent execution; a columnar storage engine that persists frames with temporal indexing; a replay engine that reconstructs synchronized dual-pane views of UI state and agent decision waterfalls; and a search engine that enables filtering sessions by user, metadata, outcome and temporal range.

Implementation Reference

[0015] The AI DVR system may be implemented in the file `src/services/ai-dvr.js` using the classes `SessionRecorder` and `AIDVRService`. Production deployments may use ClickHouse as the columnar storage engine.

References

[1] Columnar databases store data by columns, allowing compression and efficient processing of query ranges for high-performance analytics【512259054461214†L553-L565】

[2] ClickHouse stores data by column and compresses it, enabling fast analytical queries on large datasets【354599841446093†L347-L365】